REPORT

ARTIFICIAL INTELLIGENCE (CSA1704)

Assessment Tool 1 – Industry Scenario

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INTRODUCTION

Diabetes is a chronic disease in which the body cannot properly regulate blood glucose levels. Early identification of individuals at high risk can support timely medical evaluation and lifestyle intervention. Traditional diagnosis depends heavily on clinical examination and laboratory measurements. Machine Learning can complement these processes by identifying patterns in patient data and estimating diabetes risk.

Healthcare datasets contain multiple variables that may contribute to diabetes. Examples include glucose level, BMI, age, blood pressure, cholesterol, and family history. Machine Learning algorithms can analyse these variables and learn relationships between patient characteristics and diabetes outcomes.

This project combines multiple AI approaches. First, supervised learning is used to predict whether a patient is diabetic or non-diabetic. Second, statistical learning is used for risk stratification and interpretation of predictors. Finally, Reinforcement Learning is introduced for treatment recommendation based on changes in patient health states over time.

The system is intended as a **clinical decision-support approach**. It should not replace qualified healthcare professionals or independently prescribe medical treatment.

PROBLEM STATEMENT

A healthcare company has collected patient records containing:

- Age
- Blood pressure
- Cholesterol
- Glucose level
- BMI
- Family history

The company wants to develop a predictive model for early diabetes diagnosis and an AI agent that can recommend personalised treatment actions such as:

- Diet
- Exercise
- Medication

- Monitoring

The system must address class imbalance, model evaluation, feature importance, pruning, statistical learning, and reinforcement learning.

AIM

To develop an AI-based healthcare decision-support system that predicts diabetes risk using Machine Learning and learns personalised treatment recommendations using Reinforcement Learning.

OBJECTIVES

1. To prepare and preprocess the healthcare dataset.
2. To identify important predictors of diabetes.
3. To build a Decision Tree classifier for diabetes diagnosis.
4. To handle class imbalance between diabetic and non-diabetic cases.
5. To evaluate the classifier using Accuracy, Precision, Recall, F1-Score, and False Negatives.
6. To apply post-pruning to reduce Decision Tree overfitting.
7. To develop a Logistic Regression statistical learning model.
8. To compare the Decision Tree and Logistic Regression models.
9. To identify the top three predictors of diabetes.
10. To formulate treatment recommendation as a Markov Decision Process.
11. To apply Q-Learning for treatment recommendation.
12. To consider safety, privacy, fairness, and explainability before deployment.

DATASET DESCRIPTION

The dataset contains patient-level healthcare attributes that can be used to predict diabetes.

Feature	Description
Age	Age of the patient
Blood Pressure	Patient's blood pressure measurement

Feature	Description
Cholesterol	Cholesterol level
Glucose	Blood glucose level
BMI	Body Mass Index
Family History	Presence or absence of diabetes in family
Diabetes Outcome	Target variable

The target variable is:

- **0 – Non-diabetic**
- **1 – Diabetic**

The input features are used by the machine-learning models to estimate the diabetes outcome.

METHODOLOGY

The overall methodology consists of the following stages:

Patient Dataset

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Data Cleaning

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Preprocessing

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Train-Test Split (70:30)

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Class Imbalance Handling

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Decision Tree Model

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Model Evaluation

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Post-Pruning

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Logistic Regression

↓

Feature Importance

↓

MDP Formulation

↓

Q-Learning

↓

Treatment Recommendation

DATA PREPARATION

Data Cleaning

Missing values are identified and appropriately handled. Invalid or abnormal values are also checked before training the models.

Encoding

Categorical variables such as family history are converted into numerical form.

For example:

No → 0

Yes → 1

Normalisation

Numerical features can be scaled so that variables with different ranges are comparable.

Normalisation is particularly useful for Logistic Regression and other distance/scale-sensitive algorithms. Decision Trees generally do not require normalisation.

Dataset Splitting

The dataset is divided into:

- **70% training data**
- **30% testing data**

A stratified split is preferred because it maintains the class distribution in both sets.

HANDLING CLASS IMBALANCE

Healthcare datasets may contain significantly more non-diabetic cases than diabetic cases. Training directly on such data can make the model favour the majority class.

SMOTE

SMOTE generates synthetic examples of the minority diabetic class.

SMOTE should be applied only to the training data to prevent information leakage from the test set.

Class Weighting

Class weights can also be used:

class_weight = "balanced"

This assigns greater importance to the minority class during model training.

For this healthcare problem, **SMOTE or class weighting is preferable to simple random under-sampling**, because removing genuine patient observations may discard useful information.

DECISION TREE CLASSIFICATION

A Decision Tree recursively divides the dataset according to feature values.

The tree can use either **Gini Index** or **Information Gain** to select the best split.

Gini Index

[
Gini=1-\sum p_i^2
]

A split with lower impurity is preferred.

Information Gain

$$IG=H(parent)-\sum_v\frac{|S_v|}{|S|}H(S_v)$$

The feature with the highest information gain can be selected as the root.

For a diabetes dataset, glucose level is expected to be an important splitting feature because blood glucose is strongly associated with diabetes. However, the actual root should be determined from the trained dataset.

MODEL EVALUATION

The Decision Tree is evaluated using a confusion matrix containing:

- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

Accuracy

$$Accuracy=\frac{TP+TN}{TP+TN+FP+FN}$$

Precision

$$Precision=\frac{TP}{TP+FP}$$

Recall

$$Recall=\frac{TP}{TP+FN}$$

F1-Score

$$F1=2\frac{Precision\times Recall}{Precision+Recall}$$

In this application, **Recall is particularly important** because a False Negative represents a patient with diabetes being incorrectly classified as non-diabetic.

FALSE NEGATIVE ANALYSIS

A False Negative occurs when:

Actual: Diabetic

Predicted: Non-diabetic

This can be clinically significant because the patient may not receive timely evaluation or intervention.

False negatives can be reduced by:

- Applying SMOTE.
- Using class weights.
- Adjusting the classification threshold.
- Improving feature selection.
- Hyperparameter tuning.
- Selecting a model with better diabetic-class recall.

The final system should prioritise patient safety rather than relying only on overall accuracy.

DECISION TREE PRUNING

A very deep Decision Tree may memorise training examples and suffer from overfitting.

Cost-complexity pruning is used to remove unnecessary branches.

$$R_\alpha(T)=R(T)+\alpha|T|$$

where:

- ($R(T)$) represents tree error.
- ($|T|$) represents the number of terminal nodes.

- (α) controls tree complexity.

A pruned tree is generally easier to interpret and may generalise better to unseen patients.

STATISTICAL LEARNING USING LOGISTIC REGRESSION

Logistic Regression is used because the target variable is binary.

The probability of diabetes is represented as:

$$
P(Y=1|X)=\frac{1}{1+e^{-z}}
$$

where:

$$
z=\beta_0+\beta_1X_1+\beta_2X_2+\cdots+\beta_nX_n
$$

The model produces a probability of diabetes rather than only a class label.

For example, a predicted probability of 0.80 indicates that the model estimates a relatively high risk, although the result must be interpreted using the selected clinical threshold.

FEATURE IMPORTANCE

Feature importance is analysed to determine which patient characteristics contribute most strongly to diabetes prediction.

Potential important predictors include:

1. **Glucose Level**
2. **BMI**
3. **Age**

Other variables such as blood pressure, cholesterol, and family history may also contribute.

The Decision Tree uses impurity reduction to estimate feature importance, while Logistic Regression can use the magnitude of standardised coefficients.

If both models identify similar predictors, confidence in the importance of those variables increases. Differences can occur because the two models represent relationships differently.

REINFORCEMENT LEARNING

Treatment recommendation is formulated as a **Markov Decision Process (MDP)**.

[
MDP=(S,A,P,R,\gamma)
]

States

Example patient health states:

- Low Risk
- Moderate Risk
- High Risk
- Controlled

Actions

- Diet
- Exercise
- Medication
- Monitor

Reward

The reward represents improvement in patient health.

Example:

Outcome	Reward
Significant improvement	+10
Moderate improvement	+5
Small improvement	+1
No improvement	0
Deterioration	-5

These values are illustrative. Real clinical reward functions must be based on validated medical outcomes.

Q-LEARNING

Q-Learning is a model-free Reinforcement Learning algorithm that learns the value of taking an action in a particular state.

The update equation is:

$$
Q(s,a)\leftarrow Q(s,a)+
\alpha[r+\gamma\max Q(s',a')-Q(s,a)]
$$

where:

- (s) = current state
- (a) = current action
- (r) = reward
- (s') = next state
- (α) = learning rate
- (γ) = discount factor

The agent repeatedly interacts with the simulated patient environment and updates its Q-table.

Example:

State	Diet	Exercise	Medication	Monitor
Low Risk	3.5	2.5	0.5	2.0
Moderate Risk	3.0	4.2	2.5	1.5
High Risk	1.0	2.0	5.5	3.0
Controlled	1.5	3.0	1.0	4.0

The highest Q-value in each row represents the currently preferred action.

CONVERGENCE

Q-Learning training can be considered stable when the change in Q-values becomes sufficiently small:

$$
\max |Q_{new}-Q_{old}|<\epsilon
$$

For example:

$$
\epsilon=0.001
$$

can be selected as a convergence threshold.

At least five patient episodes should be used for the practical experiment, with Q-table updates demonstrated for multiple state-action pairs across iterations.

FINAL POLICY

The learned policy is obtained using:

$$
\pi(s)=\arg\max_a Q(s,a)
$$

An illustrative policy is:

Patient State	Recommended Action
Low Risk	Diet
Moderate Risk	Exercise
High Risk	Medication
Controlled	Monitor

This table is only an example of the RL output. Actual medical recommendations must be generated from clinically validated patient-transition data and must remain subject to clinician approval.

ETHICAL CONSIDERATIONS

Patient Safety

AI-generated treatment recommendations must not be treated as autonomous medical prescriptions.

Bias

The training data should represent the intended patient population. Performance should be evaluated across relevant demographic and clinical groups.

Explainability

The system should provide understandable reasons for predictions and recommendations.

Privacy

Patient information is sensitive and must be protected through appropriate security controls, access restrictions, and data protection mechanisms.

Human Oversight

Qualified healthcare professionals should review important predictions and treatment recommendations.

Accountability

The system should maintain appropriate records of model versions, predictions, recommendations, and human decisions.

EXPECTED RESULTS

The expected outcome of the project is an integrated AI system capable of:

- Predicting diabetes risk.
- Identifying important diabetes predictors.
- Handling imbalanced medical data.
- Providing Decision Tree-based interpretable predictions.
- Comparing Decision Tree and Logistic Regression performance.
- Identifying false-negative cases.
- Reducing overfitting through pruning.
- Learning treatment policies using Q-Learning.

- Supporting personalised healthcare decision-making.

The actual Accuracy, Precision, Recall, F1-Score, AUC-ROC, feature rankings, and Q-values should be reported from the executed implementation rather than using fabricated values.

APPLICATIONS

The proposed system can support:

1. Early diabetes risk screening.
2. Patient risk stratification.
3. Clinical decision support.
4. Personalised lifestyle recommendations.
5. Continuous patient monitoring.
6. Identification of high-risk patients.
7. Healthcare analytics and research.

LIMITATIONS

1. Model performance depends on dataset quality.
2. Missing or biased data can affect predictions.
3. Machine Learning predictions are not guaranteed to be medically correct.
4. Q-Learning requires reliable patient-state transition information.
5. Treatment recommendations require clinical validation.
6. Privacy and security must be maintained throughout the system lifecycle.
7. The model should not replace professional medical diagnosis.

FUTURE ENHANCEMENTS

Future versions can include:

- Larger and more diverse healthcare datasets.
- Real-time patient monitoring.
- Wearable-device data integration.
- Explainable AI techniques such as SHAP.
- Deep Learning models for complex patterns.

- Personalised risk prediction.
- Clinician feedback integration.
- More advanced safe Reinforcement Learning methods.
- Continuous model monitoring and bias detection.

CONCLUSION

This project demonstrates how Artificial Intelligence can be applied to diabetes diagnosis and personalised treatment decision support. The Decision Tree provides an interpretable supervised-learning approach for classification, while Logistic Regression provides a statistical method for estimating diabetes probability and analysing predictor effects.

Class imbalance can be addressed using SMOTE or class weighting, while post-pruning can improve the generalisation and interpretability of Decision Trees. Performance should be assessed using multiple metrics, with particular attention to Recall and False Negatives because missed diabetic cases may have serious consequences.

The Reinforcement Learning component models treatment recommendation as a Markov Decision Process and uses Q-Learning to learn action policies from patient-state transitions. Although this approach can support personalised decision-making, medical treatment recommendations require strong safety controls, clinical validation, privacy protection, fairness assessment, explainability, and human oversight.

Therefore, the proposed system should be implemented as an **AI-assisted clinical decision-support system**, with final medical decisions remaining under the responsibility of qualified healthcare professionals.

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